

Drawing View Detection on Engineering Sheets

Comparison of Approaches Evaluated - Vector Methods, Grounding DINO, and VLM-Assisted Pipelines

1. Objective

This document compares the approaches evaluated for detecting drawing views/boundaries and their associated titles on engineering sheets (e.g., front, top, side, section, or isometric views within a single sheet). It summarizes the method, results, and limitations of each approach and proposes a recommended direction going forward.

2. Problem Context

Engineering drawing sheets typically contain one or more views (drawings) of a part or assembly, each accompanied by a title block or label (e.g., referencing scale, view name, or archival reference text). Sheets do not follow a single standard layout, and the number of drawings per sheet varies. The task is to reliably detect:

- The boundary/region of each individual drawing on a sheet
- The title associated with each drawing

Four approaches were evaluated, summarized below and detailed in Section 4.

3. Summary Comparison

Criteria	Approach 1 Vector Clustering	Approach 2 Grounding DINO	Approach 3 Qwen + Grounding DINO	Approach 4 Vector Title + Tuned DINO (Proposed)
Core method	Classical CV / geometric vector clustering	Zero-shot open-vocabulary detection for drawing boundary	VLM (Qwen) for title text + Grounding DINO for boundary	Vector/rule-based title extraction (scale & arch-ref text) + threshold-tuned Grounding DINO
Semantic understanding	None	Limited (visual grounding only)	Yes (via Qwen)	Rule-based text cues (no VLM) — lightweight semantic proxy

Criteria	Approach 1 Vector Clustering	Approach 2 Grounding DINO	Approach 3 Qwen + Grounding DINO	Approach 4 Vector Title + Tuned DINO (Proposed)
Generalization across sheet formats	Poor — breaks on non-standard layouts	Moderate — needs fixed thresholds per case	Moderate — limited by DINO false positives	Improved — threshold tuned per drawing count + single/tiled inference
Compute / inference cost	Low	Low–Medium	High (Qwen inference heavy)	Low–Medium (no VLM dependency)
Inference speed	Fast	Fast	Slow — hard to test at scale	Faster than Approach 3 (VLM removed)
Key limitation	No standard format across drawings; not generalizable	Requires manual/fixed threshold tuning; not adaptive	Heavy load, long test cycles; DINO gives high false positives	Requires well-defined scale/arch-ref text conventions; tuning still manual per sheet type
Accuracy / Precision / Recall	[ADD METRICS]	[ADD METRICS]	[ADD METRICS]	[ADD METRICS]
Status	Rejected — not generalized	Superseded	Superseded — too slow, high false positives	Proposed — latest approach

Note: Quantitative metrics (accuracy / precision / recall / IoU / inference time) are placeholders and to be filled in once benchmarking is finalized.

4. Detailed Approach Analysis

Approach 1: CV / Vector-Based Clustering

REJECTED

Method: Classical computer vision using vector-based clustering to extract structural information directly from drawing geometry (lines, curves, contours).

How it works: Extracts geometric/vector structure of the drawing and clusters elements based on spatial and structural proximity to infer drawing boundaries.

Strengths: No model training or inference cost; works purely on structural/geometric cues; fast and lightweight.

Limitations: No standard sheet format exists across drawings, so clustering rules do not generalize. The approach has no semantic understanding of the content (cannot distinguish a title block from a drawing, or reason about context), which fundamentally limits generalization.

Outcome: Not viable as a generalized, standalone solution.

Approach 2: Grounding DINO for Drawing Boundary Detection

SUPERSEDED

Method: Open-vocabulary object detection model (Grounding DINO) used to directly detect drawing boundaries on the sheet via text-prompted detection.

How it works: Grounding DINO is prompted with a text query describing a drawing/view, and its detection boxes are used as candidate drawing boundaries, filtered using a fixed confidence threshold.

Strengths: Works out-of-the-box without task-specific training; reasonable results on some sheet layouts.

Limitations: Works only partially, and only with a fixed threshold — performance is inconsistent across different sheet types and requires manual/fixed thresholding rather than adapting to sheet content. Needs further fine-tuning to be reliable.

Outcome: Usable as a component, but not sufficient alone; carried forward into Approaches 3 and 4 with refinements.

Approach 3: Qwen (Titles) + Grounding DINO (Boundary)

SUPERSEDED

Method: Combines a Vision-Language Model (Qwen) for identifying drawing titles with Grounding DINO for drawing boundary detection.

How it works: Qwen is used to identify/read titles within the sheet, supporting single-image and tiled inference modes. Grounding DINO independently detects drawing boundaries; outputs are combined.

Strengths: Qwen brings genuine semantic/text understanding for identifying titles, an improvement over pure geometric methods.

Limitations: Qwen's inference load is heavy, making it slow and difficult to test at scale (both single and tiled inference modes were impacted). Grounding DINO continued to produce a high rate of false positives, limiting overall pipeline accuracy.

Outcome: Semantically more capable than Approaches 1–2, but impractical due to inference cost/time and boundary false positives.

Approach 4: Vector-Based Title Extraction + Threshold-Tuned Grounding DINO (Proposed)

PROPOSED — LATEST APPROACH

Method: Replaces the VLM (Qwen) with a lightweight vector/rule-based extraction method for titles, based on scale and architecture-reference text cues, combined with Grounding DINO for boundary detection — with detection thresholds tuned based on the number of drawings present on a sheet.

How it works: Titles are extracted via vector-based text/structure extraction keyed on scale and arch-reference text, without invoking any VLM. Grounding DINO's confidence threshold is dynamically tuned based on drawing count per sheet, using both single and tiled inference modes for different drawing types.

Strengths: Removes the heavy VLM dependency (faster, lighter than Approach 3); adapts detection thresholds instead of using a fixed value, addressing Approach 2's key weakness; supports both single and tiled inference to handle varying drawing types/sizes.

Limitations: Threshold tuning by drawing count still requires calibration per sheet type; relies on consistent presence of scale/arch-ref text conventions for title extraction to work reliably. Full quantitative validation still pending.

Outcome: Current best-performing and most practical approach; recommended as the proposed solution, pending final benchmarking.

5. Recommendation

Approach 4 (Vector-Based Title Extraction + Threshold-Tuned Grounding DINO) is proposed as the go-forward solution:

- Eliminates the heavy VLM (Qwen) inference bottleneck identified in Approach 3, improving speed and testability.
- Improves on Approach 2 by tuning detection thresholds based on the number of drawings per sheet rather than using a single fixed threshold.
- Retains a lightweight semantic cue (scale/arch-reference text) for title extraction without the overhead of a VLM.
- Supports both single and tiled inference to accommodate different drawing types and sheet densities.

This approach should be finalized with full quantitative benchmarking (accuracy, precision/recall, IoU, inference time) across a representative sample of sheet formats before sign-off.